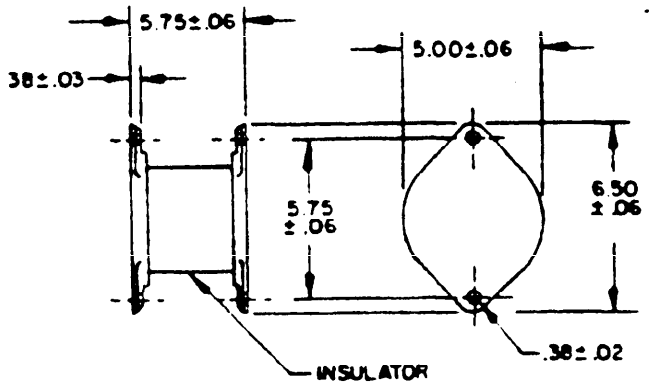
MILITARY SPECIFICATION SHEET
CAPACITORS, FIXED MICA DIELECTRIC
STYLES CM90, CM91, AND CM93 1/

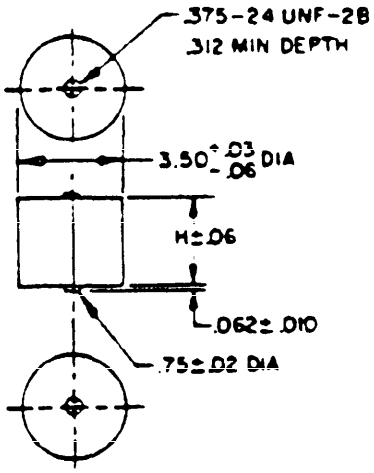
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.

STYLE CM90 IS INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION.



STYLE CM90

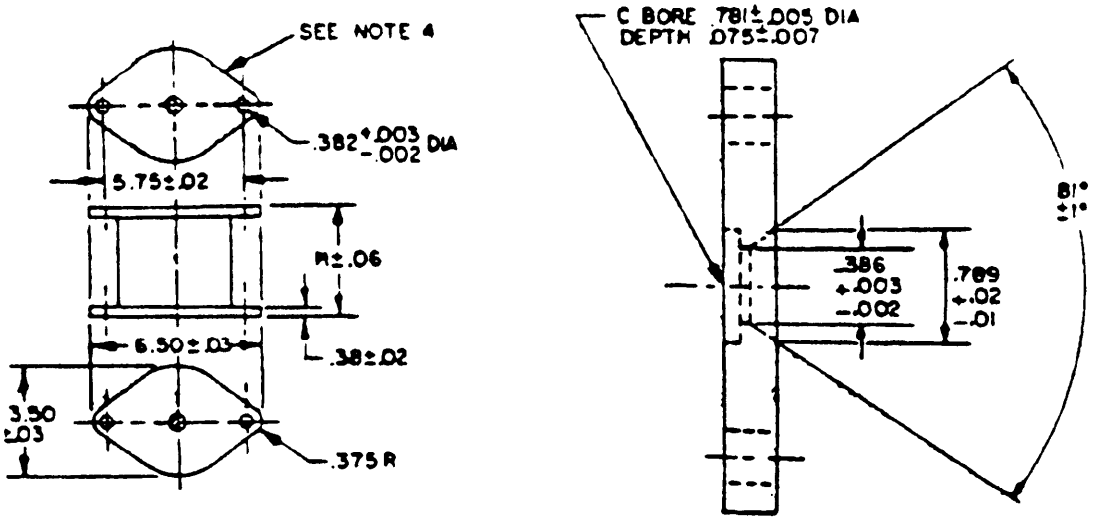


STYLE CM91

1/ CM92 deleted.

FSC 5910

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STYLE CM93

INCHES	MM
.002	.05
.003	.08
.01	.25
.010	.25
.02	.51
.03	.76
.06	1.52
.062	1.57
.312	7.92
.375	9.53
.38	9.65
.382	9.70
.386	9.80
.75	19.05
.781	19.84
.789	20.04
3.50	88.90
5.00	127.00
5.75	146.05
6.50	165.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005(.13 mm) for three place decimals.
4. Brackets (flanges) shall be removable.
5. Hardware to be plated against corrosion.
6. Approximate weight of CM93 is 8 pounds (3.63 kg).

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REQUIREMENTS:

Dimensions: See figure.

Material: Bracket (flange) - Aluminum per QQ-A-250/4 (T3-temper).

Construction: CM90 - Molded case.
CM91 and CM93 - Cast epoxy case.

Capacitance value: See table I.

Capacitance tolerance: ±5% (symbol J).

Peak working voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
M	-55 to +70°C
O	-55 to +125°C

Vibration grade: Grade 1 (10-55 Hz).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
B	Not rated	Not rated
D	-100 to +100	±(0.3 percent + 0.1 pF)
E	-20 to +100	±(0.1 percent + 0.1 pF)
F	0 to +70	±(0.05 percent + 0.1 pF)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
AC sine-wave potential rms equal to peak working voltage at 100 cycles or less for not less than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202, condition B (50,000 ft)
Test potential - 30 percent of rated peak voltage.

Insulation resistance:
Method 302 of MIL-STD-202, condition A
At 25°C:
(CM90) - 7,500 megohms minimum.
(CM91 and CM93) - 10,000 megohms minimum.
At 70°C:
(CM90) - 0 to 5,000 pF -- 2,000 megohms minimum.
greater than 5,000 pF -- 10 megohm microfarads.
At 125°C:
(CM91 and CM93) - 0 to 5,000 pF -- 3,000 megohms minimum;
greater than 5,000 pF -- 15 megohm microfarads.

Dissipation factor, maximum - See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Current rating:
Passage of rated current shall not cause the temperature of the capacitor body's external surface to rise above the ambient temperature by more than 15°C for style CM90 and not more than 40°C for styles CM91 and CM93.

Vibration:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high temperature shall be as follows:
Symbol M - +70°C
Symbol O - +125°C

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 3,000 megohms minimum
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pF, whichever is greater.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G(pk)).

Moisture resistance.
Method 106 of MIL-STD-202:
Insulation resistance - At least 30 percent of initial requirement
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B.
±2 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

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Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at -55 $^{+0}_{-2}$°C immediately followed by qualification or performance test, as applicable.
Qualification test - 1,000 hours at the applicable high test temperature with rated current at 1 MHz.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±8 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±3 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Performance check - 250 hours at conditions specified for qualification test.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.
Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 750 hours and meet the requirements for the qualification test. (This test not applicable to styles CM91 and CM93.)

Qualification inspection:
Qualification inspection (Group E) is required.

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TABLE I. Styles CM90, CM91, and CM93

Type designation 1/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM90-W101JM1	100	30,000	7.5	5.1	2.0	0.68	B, D, F
CM90-W111JM1	110	30,000	8.2	5.1	2.2	0.82	
CM90-W121JM1	120	30,000	8.2	5.6	2.4	1.0	
CM90-W131JM1	130	30,000	9.1	6.2	2.7	1.1	
CM90-W151JM1	150	30,000	9.1	6.2	3.0	1.3	
CM90-W161JM1	160	30,000	9.1	6.8	3.3	1.5	
CM90-W181JM1	180	30,000	10	6.8	3.6	1.6	
CM90-W201JM1	200	30,000	10	7.5	3.9	1.8	
CM90-W221JM1	220	30,000	11	7.5	3.9	2.0	
CM90-W241JM1	240	30,000	11	8.2	4.3	2.0	
CM90-W271JM1	270	30,000	11	9.1	4.7	2.2	
CM90-W301JM1	300	30,000	12	9.1	5.1	2.4	
CM90-W331JM1	330	30,000	12	9.1	5.1	2.4	
CM90-W361JM1	360	30,000	12	10	5.6	2.7	
CM90-W391JM1	390	30,000	13	11	6.2	3.0	
CM90-W431JM1	430	30,000	13	11	6.2	3.0	
CM90-W471JM1	470	30,000	13	11	6.8	3.3	
CM90-W511JM1	510	30,000	13	12	6.8	3.3	
CM90-W561JM1	560	30,000	15	13	7.5	3.6	
CM90-W621JM1	620	30,000	15	13	8.2	3.9	
CM90-W681JM1	680	30,000	15	13	8.2	4.3	
CM90-W751JM1	750	30,000	16	15	8.2	4.3	
CM90-W821JM1	820	30,000	16	15	9.1	4.7	
CM90-W911JM1	910	30,000	16	15	10	5.1	
CM90-W102JM1	1,000	30,000	18	16	10	5.1	
CM90-V112JM1	1,100	25,000	18	18	11	5.6	
CM90-V122JM1	1,200	25,000	18	18	11	5.6	
CM90-V132JM1	1,300	25,000	18	18	12	6.2	
CM90-V152JM1	1,500	25,000	20	20	12	6.2	
CM90-V162JM1	1,600	25,000	20	20	13	6.8	
CM90-V182JM1	1,800	25,000	20	22	13	7.5	
CM90-U202JM1	2,000	20,000	22	22	15	7.5	
CM90-U222JM1	2,200	20,000	22	24	15	8.2	
CM90-U242JM1	2,400	20,000	22	24	16	8.2	
CM90-U272JM1	2,700	20,000	22	24	16	9.1	
CM90-U302JM1	3,000	20,000	24	27	18	9.1	
CM90-U332JM1	3,300	20,000	24	27	18	10	
CM90-U362JM1	3,600	20,000	24	27	18	10	
CM90-U392JM1	3,900	20,000	24	30	20	11	
CM90-T432JM1	4,300	15,000	27	30	20	11	
CM90-T472JM1	4,700	15,000	27	30	22	12	
CM90-T512JM1	5,100	15,000	27	33	22	12	
CM90-T562JM1	5,600	15,000	27	36	22	13	
CM90-T622JM1	6,200	15,000	27	36	24	15	
CM90-T682JM1	6,800	15,000	27	36	24	15	
CM90-T752JM1	7,500	15,000	30	36	27	16	
CM90-S822JM1	8,200	12,000	30	39	27	16	
CM90-S912JM1	9,100	12,000	30	39	27	16	
CM90-R103JM1	10,000	10,000	30	43	30	18	
CM90-R113JM1	11,000	10,000	30	43	30	18	
CM90-Q123JM1	12,000	8,000	30	43	30	20	
CM90-Q133JM1	13,000	8,000	30	47	30	20	
CM90-Q153JM1	15,000	8,000	30	47	33	20	
CM90-Q163JM1	16,000	8,000	30	47	33	22	B, D, F

1/ See [illegible]

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TABLE I. Styles CM90, CM91, and CM93 - Continued

Type designation 1/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies - (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM90-P183JM1	18,000	6,000	30	51	36	22	B, D, F
CM90-P203JM1	20,000	6,000	30	51	36	24	
CM90-P223JM1	22,000	6,000	30	51	36	24	
CM90-P243JM1	24,000	6,000	30	56	39	27	
CM90-P273JM1	27,000	6,000	30	56	39	27	
CM90-P303JM1	30,000	6,000	30	56	39	27	
CM90-N333JM1	33,000	5,000	30	56	39	27	
CM90-N363JM1	36,000	5,000	30	56	39	30	
CM90-N393JM1	39,000	5,000	30	56	39	30	
CM90-N433JM1	43,000	5,000	30	56	39	30	
CM90-N473JM1	47,000	5,000	30	56	39	30	
CM90-N513JM1	51,000	5,000	30	56	39	33	
CM90-N563JM1	56,000	5,000	30	56	39	33	
CM90-N623JM1	62,000	5,000	30	56	39	33	
CM90-M683JM1	68,000	4,000	30	56	39	36	
CM90-M753JM1	75,000	4,000	30	62	43	36	
CM90-L823JM1	82,000	3,000	30	62	43	39	
CM90-L913JM1	91,000	3,000	30	62	43	39	
CM90-L104JM1	100,000	3,000	30	62	43	39	B, D, F

Type designation 2/	Capacitance (pF)	Rated working voltage (volts)	"H" case height (inches)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM9--W101JO1	100	30,000	4.00	7.5	5.1	2.6	0.98	D, E
CM9--W111JO1	110	30,000	4.00	8.2	5.4	2.2	0.82	
CM9--W121JO1	120	30,000	4.00	8.2	5.6	2.4	1.0	
CM9--W131JO1	130	30,000	4.00	9.1	6.2	2.7	1.1	
CM9--W151JO1	150	30,000	4.00	9.1	6.2	3.0	1.3	
CM9--W161JO1	160	30,000	4.00	9.1	6.8	3.3	1.5	
CM9--W181JO1	180	30,000	4.00	10	6.8	3.6	1.6	
CM9--W201JO1	200	30,000	4.00	10	7.5	3.9	1.8	
CM9--W221JO1	220	30,000	4.00	11	7.5	3.9	2.0	
CM9--W241JO1	240	30,000	4.00	11	8.2	4.3	2.0	
CM9--W271JO1	270	30,000	4.00	11	9.1	4.7	2.2	
CM9--W301JO1	300	30,000	4.00	12	9.1	5.1	2.4	
CM9--W331JO1	330	30,000	4.00	12	9.1	5.1	2.4	
CM9--W361JO1	360	30,000	4.00	12	10	5.6	2.7	
CM9--W391JO1	390	30,000	4.00	13	11	6.2	3.0	
CM9--W431JO1	430	30,000	4.00	13	11	6.2	3.0	
CM9--W471JO1	470	30,000	4.00	13	11	6.8	3.3	
CM9--W511JO1	510	30,000	4.00	13	12	6.8	3.3	
CM9--W561JO1	560	30,000	4.00	15	13	7.5	3.6	
CM9--W621JO1	620	30,000	4.00	15	13	8.2	3.9	
CM9--W681JO1	680	30,000	4.50	15	13	8.2	4.3	
CM9--W751JO1	750	30,000	4.50	16	15	8.2	4.3	
CM9--W821JO1	820	30,000	4.50	16	15	9.1	4.7	
CM9--W911JO1	910	30,000	4.50	16	15	10	5.1	
CM9--W102JO1	1,000	30,000	4.50	18	16	10	5.1	
CM9--V112JO1	1,100	25,000	4.50	18	18	11	5.6	
CM9--V122JO1	1,200	25,000	4.50	18	18	11	5.6	
CM9--V132JO1	1,300	25,000	4.50	18	18	12	6.2	
CM9--V152JO1	1,500	25,000	5.00	20	20	12	6.2	
CM9--V162JO1	1,600	25,000	5.00	20	20	13	6.8	
CM9--V182JO1	1,800	25,000	5.00	20	22	13	7.5	
CM9--U202JO1	2,000	20,000	4.50	22	22	15	7.5	
CM9--U222JO1	2,200	20,000	4.50	22	24	15	8.2	D, E

See footnotes at end of table

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TABLE I Styles CM90, CM91, and CM93 - Continued.

Type designation 2/	Capacitance (pF)	Rated working voltage (volts)	"E" Case height (inches)	Rated current at frequencies (rms amps)				Characteristic
				3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	
CM9--U242JO1	2,400	20,000	4.50	22	24	16	8.2	D, E
CM9--U272JO1	2,700	20,000	5.00	22	24	16	9.1	
CM9--U302JO1	3,000	20,000	5.00	24	27	18	9.1	
CM9--U332JO1	3,300	20,000	5.00	24	27	18	10	
CM9--U362JO1	3,600	20,000	5.00	24	27	18	10	
CM9--U392JO1	3,900	20,000	5.00	24	30	20	11	
CM9--T432JO1	4,300	15,000	4.50	27	30	20	11	
CM9--T472JO1	4,700	15,000	4.50	27	30	22	12	
CM9--T512JO1	5,100	15,000	4.50	27	33	22	12	
CM9--T562JO1	5,600	15,000	5.00	27	36	22	13	
CM9--T602JO1	6,000	15,000	5.00	27	36	24	15	
CM9--T622JO1	6,200	15,000	5.00	27	36	24	15	
CM9--T682JO1	6,800	15,000	5.00	27	36	24	15	
CM9--T752JO1	7,600	15,000	5.00	30	36	27	16	
CM9--8822JO1	8,200	12,000	5.00	30	39	27	16	
CM9--8912JO1	9,100	12,000	5.00	30	39	27	16	
CM9--R103JO1	10,000	10,000	5.00	30	43	30	18	
CM9--R113JO1	11,000	10,000	5.00	30	43	30	18	
CM9--Q123JO1	12,000	8,000	4.50	30	43	30	20	
CM9--Q133JO1	13,000	8,000	4.50	30	47	30	20	
CM9--Q153JO1	15,000	8,000	5.00	30	47	33	20	
CM9--Q163JO1	16,000	8,000	5.00	30	47	33	22	
CM9--P183JO1	18,000	6,000	4.00	30	51	36	22	
CM9--P203JO1	20,000	6,000	4.00	30	51	36	24	
CM9--P223JO1	22,000	6,000	4.00	30	51	36	24	
CM9--P243JO1	24,000	6,000	4.50	30	56	39	27	
CM9--P273JO1	27,000	6,000	4.50	30	56	39	27	
CM9--P303JO1	30,000	6,000	5.00	30	56	39	27	
CM9--N333JO1	33,000	5,000	4.50	30	56	39	27	
CM9--N363JO1	36,000	5,000	4.50	30	56	39	30	
CM9--N393JO1	39,000	5,000	5.00	30	56	39	30	
CM9--N433JO1	43,000	5,000	5.00	30	56	39	30	
CM9--N473JO1	47,000	5,000	5.00	30	56	39	30	
CM9--N513JO1	51,000	5,000	5.00	30	56	39	33	
CM9--N563JO1	56,000	5,000	5.00	30	56	39	33	
CM9--N623JO1	62,000	5,000	5.00	30	56	39	33	
CM9--M683JO1	68,000	4,000	5.00	30	56	39	36	
CM9--M753JO1	75,000	4,000	5.00	30	62	43	36	
CM9--L823JO1	82,000	3,000	4.00	30	62	43	39	
CM9--L913JO1	91,000	3,000	4.50	30	62	43	39	
CM9--L104JO1	100,000	3,000	4.50	30	62	43	39	D, E

1/ For style CM90, the complete type designation will include an additional symbol to indicate characteristic.

2/ For styles CM91 and CM93 the complete type designation will include additional symbols to indicate style and characteristic.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - AS, OS, EC, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - MC, CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1047-3)

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c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one) ☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify) _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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